A SUBSPACE MODEL OF SPEECH-WATERMARK SURVIVAL UNDER ANALYSIS–RESYNTHESIS

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ABSTRACT

Speech watermarks are widely reported to fail when audio is regenerated by a vocoder or neural codec. We give this phenomenon a restricted but precise model: a blind launderer is an *analysis–resynthesis channel* $\mathcal{W} = \mathcal{S} \circ \mathcal{A}$. At full generality we prove only what is true: a data-processing inequality (laundering cannot increase the Chernoff detection exponent), and an *exact-erasure condition* — if embedding leaves the analysis output unchanged a.s., the laundered laws coincide and no detector beats chance. In a linear-Gaussian surrogate we obtain closed forms: the channel transmits exactly the equivalence class of the perturbation modulo $\ker \mathcal{A}$, with surviving exponent $\frac{1}{8}\|P\delta\|^2$; a Gaussian *achievable lower bound* R_{LB} (not a capacity) covers blind embedders. The model's empirical content is a measurable, channel-relative *survival predictor*: each attacker's own analysis sensitivity. On LibriSpeech (1000 speaker-disjoint test clips, thresholds from 7157 independent calibration clips), across five resynthesis families (spectral inversion, a neural vocoder, and three neural codecs), the predictor tracks held-out post-attack separability within every attacker (pooled Spearman 0.72, perm. $p < 10^{-3}$), while SNR and spectral-centroid controls fail. Deployed watermarks (AudioSeal, WavMark, SilentCipher), each at its native transparent strength, lose their calibrated operating point under nearly every lossy channel, in modes we keep separate: *erasure* (separability itself collapses), *calibration failure* (separability remains but the fixed threshold's false-alarm rate explodes on attacked clean audio), and *graceful degradation* (recall falls but detection still works).

***Index Terms*—** audio watermarking, generative AI, provenance, data-processing inequality, hypothesis testing

1. INTRODUCTION

Provenance watermarking embeds a key-dependent signal in speech so a detector can later flag synthetic or traced audio [1, 2, 3]. A consistent empirical finding is fragility to *regeneration*: passing watermarked audio through a vocoder, neural codec, or voice conversion drives deployed detectors toward failure [4, 5]. The community's response is to embed in representations that resynthesis preserves — codec latents (Latent-Mark [6]), speech-feature-aligned marks [7], speaker-latents robust to zero-shot VC [8], and semantic image watermarks before them [9, 10, 11]. The *invariant-alignment idea is thus established practice, not our contribution.*

What the literature lacks is a stated model in which the collapse and its exceptions can be derived and *predicted*. Empirical papers report which schemes die under which attacks; the pattern — AudioSeal survives EnCodec yet fails under mel-inversion; marks with more energy in preserved components survive longer — is left as observation. This paper supplies the missing object: a channel model with exactly-scoped guarantees, and a *measurable* quantity that predicts survival out-of-sample.

Contributions. (i) A general proposition for blind analysis–resynthesis channels: a Chernoff data-processing inequality, and a sufficient condition for *exact* erasure (§2). Nothing stronger is claimed at this generality. (ii) A linear-Gaussian surrogate where everything is closed-form: the channel transmits the perturbation's class in $\mathbb{R}^n/\ker \mathcal{A}$; mixed perturbations survive *partially* with exponent $\frac{1}{8}\|P\delta\|^2$ (§3). (iii) A Gaussian achievable lower bound R_{LB} for blind embedders — explicitly *not* a watermarking capacity: informed embedding [12, 13] can exceed it and no rate converse is claimed (§3). (iv) A channel-relative survival predictor — each attacker's own analysis sensitivity — validated on *held-out, speaker-disjoint* data across five resynthesis families against competing predictors, with permutation controls (§4). (v) A calibrated operational evaluation of deployed watermarks, separating loss of *separability* (oriented AUC) from loss of the *calibrated operating point* (TPR and *achieved* FPR at a threshold fixed on an independent calibration set) — distinctions routinely conflated (§4).

2. THE ANALYSIS–RESYNTHESIS CHANNEL

A watermark embedder E_κ maps a host $x \sim P_0$ to $E_\kappa(x)$ (additively, $E_\kappa(x) = x + \delta_\kappa$), giving the watermarked law P_1. A detector knowing κ tests $H_0 : P_0$ vs $H_1 : P_1$; with N i.i.d. looks the optimal Bayes error decays as $e^{-N\,C(P_0,P_1)}$, C the Chernoff information [14, 15]. A *blind launderer* is

$$\mathcal{W} \;=\; \mathcal{S} \circ \mathcal{A}, \tag{1}$$

where $\mathcal{A}$ maps a waveform to a representation z (a mel envelope, a codec latent) and $\mathcal{S}$ regenerates a waveform from z with randomness independent of the input. The attacker knows neither κ nor the scheme (adaptive, key-aware removal [16] is out of scope). Imperceptibility is a budget $\rho(\delta; x) \le D$.

Proposition 1 (general channels)**.** *For any channel $\mathcal{W}$,*

$$C(\mathcal{W}_\# P_0, \mathcal{W}_\# P_1) \;\le\; C(P_0, P_1).$$

If moreover $\mathcal{A}(E_\kappa(x)) = \mathcal{A}(x)$ holds P_0-a.s. and $\mathcal{S}$ depends on its input only through z, then $\mathcal{W}_\# P_1 = \mathcal{W}_\# P_0$; hence $C(\mathcal{W}_\# P_0, \mathcal{W}_\# P_1) = 0$ and no detector beats chance at any sample size.

Proof. The first part is the DPI for the Chernoff family: each $-\log \int p_0^{1-s} p_1^s$ is monotone under Markov kernels and the max over s preserves monotonicity [15]. For the second, $\mathcal{W}(E_\kappa(x)) = \mathcal{S}(\mathcal{A}(E_\kappa(x)), U) = \mathcal{S}(\mathcal{A}(x), U) = \mathcal{W}(x)$ a.s. with $U \perp x$, so the output laws coincide. ☐

Code, proofs, per-clip score arrays, and locked environments: `github.com/appleweiping/resynthesis-watermark-limits`.

Proposition 1 is all we assert for real nonlinear vocoders and codecs: laundering cannot help the detector, and *exactly* analysis-invariant embeddings are *exactly* erased. It supplies no constants and no prediction of partial survival. Quantitative structure requires a model.

3. LINEAR-GAUSSIAN SURROGATE

Work in whitened coordinates, $P_0 = \mathcal{N}(0, I_n)$, perceptual weighting carried by a masking metric $M \succ 0$ ($\rho(\delta) = \delta^\top M\delta$). Let $\mathcal{A} \in \mathbb{R}^{k\times n}$ have full row rank, $P = \mathcal{A}^+\mathcal{A}$ the orthogonal projector onto $\mathrm{row}(\mathcal{A})$, and let synthesis resample the complement from the clean prior:

$$\mathcal{W}(x) = Px + (I - P)w, \qquad w \sim \mathcal{N}(0, I_n) \text{ fresh.} \tag{2}$$

Theorem 1 (exact surviving shift and exponent)**.** *Under* (2)*, for a fixed additive shift* δ*:* $\mathcal{W}_\#\mathcal{N}(\delta, I) = \mathcal{N}(P\delta, I)$ *and*

$$C(\mathcal{W}_\# P_0, \mathcal{W}_\# P_1) \;=\; \tfrac{1}{8}\|P\delta\|^2. \tag{3}$$

The channel transmits exactly the equivalence class $[\delta] \in \mathbb{R}^n/\ker\mathcal{A}$: the effective perturbation is $P\delta$. Three consequences, stated carefully: $\delta \in \ker\mathcal{A}$ is erased exactly (recovering Prop. 1); *mixed* perturbations survive *partially* (exponent $\frac{1}{8}\|P\delta\|^2 > 0$ whenever $P\delta \neq 0$); and row-space-only perturbations waste no budget on erased components but are *not* the unique surviving set. The budget-optimal exponent is $\max_{\delta^\top M\delta \le D} \frac{1}{8}\|P\delta\|^2 = \frac{D}{8}\,\lambda_{\max}(P, M)$, at the top generalized eigenvector of (P, M).

A rate lower bound (not capacity). Writing surviving shifts in an orthonormal basis of $\mathrm{row}(\mathcal{A})$, a *blind* (host-uninformed, Gaussian codebook) embedder facing a κ-aware but host-blind decoder achieves any rate below

$$R_{\mathrm{LB}} = \max_{Q\succeq 0,\ \mathrm{tr}(M_{\mathrm{row}}Q)\le D} \tfrac{1}{2}\log\det(I + Q) \tag{4}$$

nats per invariant coordinate (host as unit-variance interference; water-filling over M_{row}'s eigenmodes). R_{LB} is an *achievable lower bound for that restricted class*: informed (host-aware) embedding in the Costa/Moulin–O'Sullivan sense [12, 13] can exceed it, and we prove no rate converse. A nullspace embedding has $R_{\mathrm{LB}} = 0$ after $\mathcal{W}$.

From surrogate to real channels. For a real attacker $\mathcal{W}$ the model suggests a *measurable* channel-relative predictor: the attacker's own analysis sensitivity

$$s_{\mathcal{W}}(\delta; x) = \|\mathcal{A}_{\mathcal{W}}(x + \delta) - \mathcal{A}_{\mathcal{W}}(x)\|/\|\delta\|, \tag{5}$$

computed with *that* attacker's $\mathcal{A}_{\mathcal{W}}$ (a mel front-end, a codec encoder). Whether $s_{\mathcal{W}}$ predicts post-attack separability is an *empirical hypothesis* — the surrogate motivates it but does not prove it. Testing it out-of-sample is E2's job; a single mel fraction is never reused across attackers with different $\mathcal{A}_{\mathcal{W}}$.

4. EXPERIMENTS

Setup. LibriSpeech 16 kHz: test = 1000 speaker-stratified 4 s voice-active clips from test-clean (40 speakers); detector thresholds come *only* from 7157 independent calibration clips (dev-clean, dev-other, and test-other, all speaker-disjoint from the test-clean evaluation set); the predictor's mapping is fit on a dev-clean fit split and evaluated on test-clean only. Ten attackers: STFT-GL (near-lossless control),

Table 1. Deployed watermarks under analysis–resynthesis at matched median PESQ. Cells: oriented AUC / TPR / *achieved* FPR at the 1%-FPR threshold fixed on the independent calibration split ($n = 7157$ clean negatives). Outcomes: *erasure* (AUC→0.5, TPR→0); *calibration failure* (AUC retained, achieved FPR explodes on attacked clean audio); *graceful degradation* (AUC and FPR retained, TPR reduced — AudioSeal/DAC).

Channel	AudioSeal	WavMark	SilentCipher
clean	1.00/1.00/0.01	1.00/1.00/0.00	0.99/0.98/0.00
STFT†	0.99/0.90/0.00	1.00/1.00/0.00	0.95/0.90/0.00
mel-inv	0.57/0.01/0.00	0.50/0.00/0.00	0.50/0.00/0.01
Vocos	0.63/0.07/0.01	0.50/0.00/0.00	0.50/0.01/0.00
Vc-EnC	0.52/0.01/0.00	0.50/0.00/0.00	0.50/0.01/0.00
EnC6	0.93/0.99/0.73	0.50/0.00/0.00	0.50/0.00/0.00
EnC3	0.81/0.97/0.80	0.50/0.00/0.00	0.50/0.00/0.01
DAC	0.93/0.64/0.01	0.50/0.00/0.00	0.50/0.01/0.01
SNAC	0.50/0.00/0.00	0.50/0.00/0.00	0.50/0.00/0.01
VC_4	0.51/0.01/0.01	0.50/0.00/0.00	0.50/0.00/0.01
VC_8	0.51/0.02/0.02	0.50/0.00/0.00	0.50/0.01/0.01

mel-GL (spectral inversion of THE unified 80-mel analysis), Vocos (neural vocoder), Vocos-EnCodec (learned reconstruction from codec tokens), EnCodec at 6/3 kbps, DAC, SNAC, and kNN-VC self voice-conversion (top-$k \in \{4, 8\}$) — content-, speaker- and prosody-preserving re-synthesis. The *predictor* study (E2) uses the five with a common analysis representation (mel-GL, Vocos, EnCodec, DAC, SNAC). All formal runs fail loudly if any attacker or baseline is missing. Scores, seeds, keys, manifests, and locked environments ship with the repository.

Constructed marks (geometry probes). W.r.t. the unified mel analysis $\mathcal{A}$ we construct, per utterance and key: a *kernel mark* (alternating projections onto quadrature ∩ consistent spectrograms — the first-order kernel of magnitude analysis) and a *row mark* (in-phase mel-pattern shifts). Construction quality is *measured*, not assumed: at the operating amplitude the kernel mark's analysis change is only $0.16\times$ that of a random direction of equal norm (residual second-order leakage), whereas the row mark's is $2.32\times$. Their detection uses a genie-aided matched-filter *probe* (the known per-utterance perturbation) — a diagnostic of channel geometry, explicitly not a deployable detector; deployed-detector conclusions come only from the baselines.

E2 — do geometric predictors predict survival? For 108 random directions (kernel/row mixtures, random in-phase) × randomized budgets, we measure $s_{\mathcal{W}}$ (5) per attacker and post-attack detectability (oriented probe AUC against host variability) in *two detector domains*: waveform correlation and a fixed mel-reading matched filter. Mappings are fit on dev-clean and evaluated on test-clean only; all statistics are within-attacker (rank-pooled), so between-attacker level differences can neither create nor mask an association. Results (Fig. 1): $s_{\mathcal{W}}$ predicts mel-domain detectability *within every attacker* (per-attacker Spearman $0.80/0.75/0.80/0.77/0.48$; pooled 0.72, permutation $p < 10^{-3}$), as does the static magnitude-change fraction (0.80) — both *geometric* quantities — while budget (SNR, -0.13) and spectral centroid (-0.27) do not. Empirically, the near-null regime behaves as the exact-erasure condition (Prop. 1) would suggest: the 31 directions with $s_{\mathcal{W}} \le 0.25\times$ median (they exist only for the mel-analysis channels — the codecs' kernels are too small to reach) deviate from chance by median 0.03, *maximum* 0.04, versus 0.09 for the rest — an observed monotonicity of the predictor, not a proof that Prop. 1 holds for these nonlinear channels. In the *waveform* domain the codecs show the *opposite* dependence ($\rho_s = -0.76$ EnCodec, -0.47 SNAC): they transmit precisely the sub-mel detail

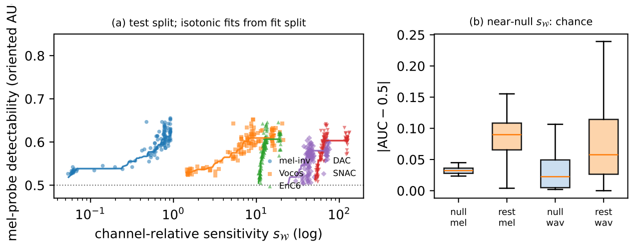

Fig. 1. E2, held-out test split. (a) Channel-relative sensitivity $s_{\mathcal{W}}$ vs. mel-probe detectability, per attacker, with isotonic fits learned on the fit split. (b) Directions in the absolute near-null regime stay at chance in both probe domains.

that mel analysis discards — survival is a property of the (channel, detector-domain) pair, not of the watermark alone.

E1 — deployed watermarks. AudioSeal [1], WavMark [2], and SilentCipher [3], each at its native full strength — all in the transparent regime but not equalized (clean-embed PESQ $4.42/4.27/4.60$; we do not force a common PESQ, which would push each scheme off its designed operating point, and we note below that AudioSeal carries the most energy). Per (baseline, attacker, key) we report oriented AUC (utterance-clustered bootstrap CIs) and the calibrated operating point — TPR *and achieved FPR* at the 1%-FPR threshold fixed on 7157 independent clean negatives (Clopper–Pearson intervals) — plus an attack-aware recalibration diagnostic (Table 1). We keep two questions separate: does *separability* survive (oriented AUC, CI vs. 0.5) and does the *calibrated operating point* survive (TPR at the fixed-threshold's achieved FPR)? All three baselines are perfect on the near-lossless STFT control. SilentCipher and WavMark lose *both* under all nine lossy channels (*erasure*: AUC $0.99 \rightarrow 0.50$, TPR $\rightarrow 0.00$, unrecoverable). AudioSeal is more varied: separability is *erased* only by SNAC and self-VC (AUC $\rightarrow 0.50$–0.51, CI covers 0.5); under mel-inversion, Vocos and Vocos-EnCodec it retains *weak* separability (AUC 0.54–0.63, CIs above 0.5) but the operating point is dead (TPR ≤ 0.07); under EnCodec it keeps *strong* separability (AUC $0.93/0.81$) yet the fixed threshold's FPR explodes ($0.01 \rightarrow 0.73/0.80$) as the codec pushes *clean* audio across the boundary (a *calibration failure*); and under DAC it merely *degrades* (AUC 0.93, FPR 0.01, recall 0.64). Survival is channel-specific in the way the subspace picture suggests — what a scheme's detector reads, relative to what each channel preserves and how it moves clean signals — and no sign/order inversion occurred in the final protocol.

E3 — multi-bit proof of concept (no rate claim). An 8-bit payload in the unified mel domain (frame-differential BPSK, fully *blind* decoder, PESQ 4.2) is carried at BER 0.10 under mel-GL and Vocos and 0.17 under EnCodec and DAC (clean 0.05), but fails under SNAC (BER 0.47); all with binomial CIs. This demonstrates the invariant sub-channel carries bits through most resynthesizers; it is *not* an achievable-rate measurement and we claim no bits/s at target reliability.

5. RELATION TO PRIOR WORK

Attack studies document the collapse [4, 5]; invariant/ latent-aligned embedding is established practice: Latent-Mark optimizes waveforms to shift codec latents with cross-codec transfer [6], Feature-Aligned marks match speech-feature distributions [7], VoiceMark targets VC-invariant speaker latents [8]; in images, semantic watermarks realize the same prescription [9, 10, 11]. Relative to all of these we add what a method paper does not: a stated channel model with exactly-scoped guarantees (Prop. 1, Thm. 1), an explicit account of what is *not* proven (R_{LB} vs capacity [13, 12, 17]), and a falsifiable, channel-relative predictor validated out-of-sample with permutation controls. Our evaluation protocol (independent calibration, oriented-vs-raw separation) addresses measurement pathologies that affect published robustness numbers.

6. SCOPE AND LIMITATIONS

The theory covers blind, non-adaptive laundering; key-aware removal [16] is out of scope. Exact erasure requires exact analysis invariance — our constructed kernel marks realize it only to first order, with measured second-order leakage at operating amplitude. R_{LB} concerns a restricted embedder class; real capacities may be higher. The surrogate's closed forms are not claimed for nonlinear generators — for those we claim only Prop. 1 and the *measured* predictive power of $s_{\mathcal{W}}$. The three deployed baselines run at native strength and are not PESQ-equalized (SI-SDR $26/37/34$ dB); AudioSeal carries the most mark energy, so its greater survival under DAC/EnCodec is partly a strength effect we do not disentangle. Survival statements are about oriented separability or calibrated operating points, never raw sub-0.5 AUC read as erasure.

7. CONCLUSION

A restricted but precise subspace model of analysis–resynthesis explains, with correctly-scoped guarantees, when speech watermarks survive laundering, and yields a measurable channel-relative predictor that generalizes across attackers and held-out speakers. Honest accounting — of what is proved, what is measured, and what deployed detectors actually do under attack — is the paper's method as much as its message.

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